Stelios Zacharioudakis

ML SYSTEMS & SOFTWARE ENGINEERING

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Software engineering for ML evaluation and deployed applications: bounded execution, APIs, data pipelines, reproducible artifacts and operational safeguards.

ENGINEERING EXPERIENCE

Former Head Engineer - Paphos Medical Association | Pro bono

Jun 2022 - Jul 2026

• Led pro bono engineering for AsklepiosMed, a donated member-services platform for the association.

• Developed full-stack web applications and administrative workflows using React, Node.js/Express and PostgreSQL.

• Maintained the platform and supported its deployment and operation on a Linux VPS.

RESEARCH

World-model learning and evaluation | Independent research, ongoing and unpublished

• Investigate imagination horizons, critic stability and policy selection with PyTorch latent world models and imagined actor-critic learning.

• Design controlled ablations, paired simulator evaluations and traceable analyses to study task-dependent outcomes and failed hypotheses.

Score-based generative models for undersampled MRI reconstruction | BSc thesis: Excellent (10/10)

• Reimplemented a published reconstruction method in PyTorch using the authors' pretrained prior; developed MRI measurement operators and a Flax-to-PyTorch checkpoint conversion.

• The thesis reports reproduction within 0.15 dB PSNR on 256 BraTS slices at 4x, 8x and 24x sampling factors; evaluates Monte Carlo uncertainty and retrospective domain shift.

Thesis: September 2026 | Evidence + independent CPU ensemble diagnostics. Retrospective research; no clinical validation.

SELECTED SOFTWARE PROJECTS

ForgeRL / ForgeBench - Built isolated coding-agent evaluation and durable budget controls. A frozen prospective pilot completed 162 episodes across 27 miniature tasks and six policies, with full traces and a separate source-derived track. One seed and six test families limit generalization; model weights remain unchanged.

MTA Scan - Built a Mapbox transit workspace with autonomous collection and 15/30-minute evaluation of feed-predicted arrival-spacing proxies. Uses chronological holdouts and baselines; early feasibility only, no validated incident claim.

Inference Systems Lab - Ran 1,536 measured requests on an RTX 4090 with pinned vLLM/Qwen versions, concurrency sweeps and direct GPU telemetry. Recorded prefix-cache comparisons and output mismatches; no lossless speedup claim.

EDUCATION

BSc in Computer Science - National and Kapodistrian University of Athens

Department of Informatics and Telecommunications | June 2026 | Degree: Very Good

TECHNICAL FOCUS

ML: Python, PyTorch, NumPy, SciPy, scikit-learn; generative models, model-based RL, uncertainty analysis and experimental evaluation.

Systems: FastAPI, Node.js/Express, React, JavaScript, SQL/PostgreSQL, Redis, Linux, Nginx, Docker, Git and automated testing.